Eos Family Magic Sheet Intensive

Workbook

V3.0.1 Rev. A

www.etcconnect.com/education

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Contact education@etcconnect.com with any comments about the workbook and support files.

Purpose of the Class

Eos Family Intensive Classes are not entry-level training courses. They are fast-paced, feature-focused sessions designed for users with an enhanced knowledge of the desk. Instructors will expect attendees to be competent in full system navigation, speedy keypad operation, move-fade philosophy, referenced data structures, and intermediate knowledge of the topics being covered. Unlike other classes, the class will move very quickly. Once created, your assignments will be shared with the class and assessed, in order to provide constructive feedback to grow your skills in the topics being covered.

Learning Objectives:

After completing this class, one should be able to:

- Explore an extensive Magic Sheet to examine advanced concepts
- Use better navigation tricks such as Tabs
- Understand the use of a 'Home' Magic Sheet or Tab
- Take advantage of the idea of 'Target' Magic Sheets or Tabs
- Construct and use a Programming Surface style Magic Sheet
- Build a Control Magic Sheet for non-programming functions and maintenance
- Overall create a more complex Magic Sheet for programming

Syntax Annotation

• **Bold**	**Browser menus**
• **[Brackets]**	**Face panel buttons**
• **{Braces}**	**Softkeys and direct selects**
• **<Angle brackets>**	**Optional keys**
• **[Next] & [Last]**	**Press & hold simultaneously**
• **«Direct Select»**	**Direct Select button press**
• **⌈MS Object⌉**	**Object on a Magic Sheet**

Help

Press and hold **[Help]** and press any key to see:

- the name of the key
- a description of what the key enables you to do
- syntax examples for using the key (if applicable)

*As with hard keys, the "press and hold **[Help]**" action can be also used with softkeys and clickable buttons*

The Manual

The manual is available on the console, Tab #100.

Click on Add-a-Tab (the {+} sign), select Manual

Hold [Tab] & press [100]

Please note that it is not available on Windows XP devices or on Mac computers but is available as a download from the web site.

About Magic Sheets

Magic Sheets are user-created graphic displays that offer customizable interactive views for showing data and programming. Magic Sheet objects can be tied to show data, such as channels and palettes, or can be representational graphics, such as scenic pieces, architecture or electrics.

Magic Sheets can be designed, edited or used in offline or from any control device online. Different views of a Magic Sheet can be stored, and Magic Sheets can be printed out like traditional Magic Sheets.

Magic Sheets are powerful for everyone in a lighting team:

- A Lighting Designer may use them to lay out channel displays in a custom or specialized way.
- An Assistant Lighting Designer may use them to track data, keep up with labels, and help the Designer keep track of moves, marks, discrete timing, etc.
- A Programmer may use them to view and interact with certain portions of the rig, or to bring forward tools to make programming quicker.
- An Electrician may use them for specialty views, rig-specific layouts, or data regarding the health of fixtures.
- Non-Technical people can interact with Magic Sheets to accomplish tasks on the system without any programming knowledge, such as archi-tainment installations.
- Other Production Team members may use Magic Sheets for indication of rig status that pertains to their area of operation.

The Magic Sheet editor – while simple in design – is powerful enough to allow a myriad of options for creating a Magic Sheet space. This book is an overview of the tools in the Magic Sheet suite – how you use those tools are nearly endless.

Capacity

- 99 Magic Sheets in a show
- Each Magic Sheet can have up to 16 views

Magic Sheet Editor Review

Magic Sheets is a tool that allows you to create a custom layout to display and to interact with your console functions in different ways.

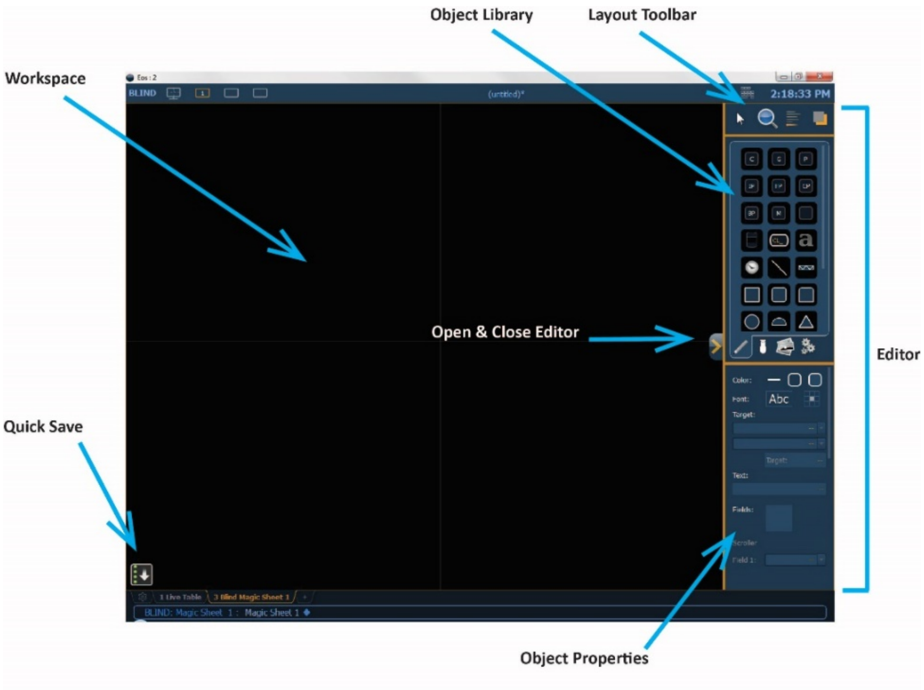
A Review of Magic Sheet Tools

[Snapshot] [101] [Enter] – MS Tabs

Add a tab next to Tab 3 (MS) on right and select another Magic Sheet tab

Create New Magic Sheet 401

Open the Editor



- Editor is divided into three sections: Layout Toolbar, Object Library, Object Properties
- Bottom left of the workspace is the Quick Save. You only have one restore point at a time to [Undo] to.
- There is an open & close editor button attached to the Editor (>). Closing the editor also creates a restore/save point

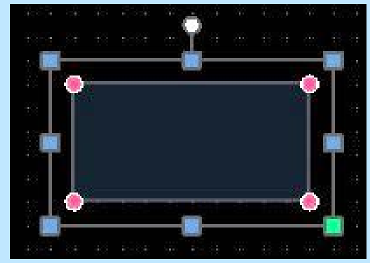
Navigating the Space: Multi-Touch

Touch with one finger	to select or drag
Touch with two fingers, move your fingers toward/away from each other	to zoom in and out
Double tap with two fingers	to zoom to all
Press and hold [Shift]	for fine control
Use three fingers to swipe upwards or to the right	to jump to the previous view
Use three fingers to swipe downwards or to the left	to jump to the next view
Tap with three fingers	to open the browser

A Review of Magic Sheet Objects

Simple Tools

Click in the Object Library on the Rectangle 7th down on the right

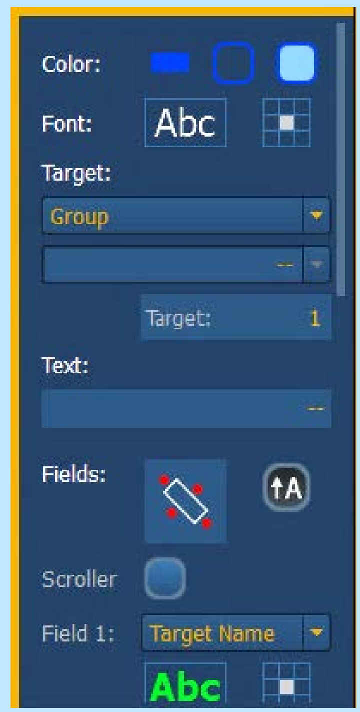


Drop it on the worksheet

- Green Handle for proportional stretch
- Blue handles for edge stretch
- White dot handle for rotate
- Pink handles for individual point move

Leave as a Rectangle, stretch it to be double the width as the height

Color Properties



- Outline width (line weight)
- Outline color
- Object fill color
  - Brightness (saturation) bar on right side
  - X is the no fill or clear.

Select an Outline width and a Fill color

Target Assignment

• Address	• Beam Palette	• Channel (default)
• Channel (By Address)	• Color Palette	• Cue
• Cue Active	• Cue Pending	• Effect
• Fader	• Focus Palette	• Group
• Intensity Palette	• Macro	• Magic Sheet
• Pixel Map	• Preset	• Response I/O Relay
• Scene	• Snapshot	• Submaster
• User		
• Console Button – second field, drop down of all console buttons		
• Softkey – select any softkey from within the software		
• Command – write command in command field, label in text field		
• Zoom - when clicked, the view will zoom in to show all objects within that object's group.		
• Selection - when clicked, all other objects within that object's group will be selected.		

Set the target as 'Group' and the target number as '102'

Field Selection

There are six fields of custom information that can be displayed. The most popular options are: Target ID, Target Name, and Label. There are also:

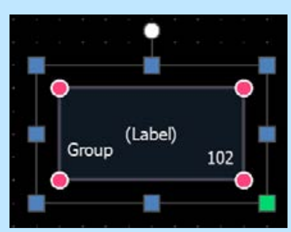
- **Abc or Font icon** - adjust the font type, size, color, style (bold, italic)
- **Justification icon** – alignment or position of the field

Make Field 1 the Target ID, bottom right justified,

Field 2 the Target Name, bottom left justified and Field 3 Label, centered

Change the fields to Interior mapping

The object might look something like the image to the right.



Select the Group 102 object and copy and paste	CTRL+C and CTRL+V
Change the Target to Color Palette and the ID to 9021	
Link to Target Color	
Move to the right of Group 102	

Align

Select both objects

Click on the Alignment tool

Align Middle

Array

Select the Color Palette 9021 object

On the Layout Toolbar, click on the Alignment Tool

Click on Create Array…

Select Rectangle

Set Columns to 5, Rows to 2 and Spacing for both at 20

Click Okay

On the Layout Toolbar, click on the Pointer

Select Quick Target tool

Change the Target to Color Palette

Change the Start to 9022 and hit okay

Renumber the rest of the top row of Color Palettes from left to right

Change the Start for the Quick Target tool to CP9031

Renumber the bottom row of Color Palettes from left to right

Don’t forget to hit Done!

Select all the Color Palettes, select the Ordering tool and group them

Close the Editor

Your Magic Sheet might look something like the image below.

INSERTING ADDITIONAL OBJECTS

Click on the Fixtures Library tab

Select the Moving Spot and drop it on the workspace below Group 102

Link Outline to Target Color

Link Fill to Target Color and Intensity

Make the object Channel 101

Fields: Exterior Shape mapping

- **Field 1: Target Id, center justified**
- **Field 2: Focus, bottom center justified**
- **Field 3: Color, bottom center justified**
- **Field 4: Beam, bottom center justified**
- **Field 5: Previous move, left middle justified**
- **Field 6: Next Move, right middle justified**

Drop a Truss over the ML

- **Resize it to fit only one fixture**
- **Select the Ordering tool and Send to Bottom**
- **Select both Channel 101 and the Truss and Align Center**

Close the Editor to show the objects

⌈101⌉ [Full] [Full]⌈Red (CP 9021)⌉

LET'S ADD SOME OTHER OBJECTS

Open the Editor

Drop a Fader to the right of the Color Palettes

Drop a Clock above Group 102

Select both the Clock and Group 102

- **Align left**

Drop a Command Line to the right of the Clock

- **Drag right side to line up with the right side of the last Color Palette**

Drop a Time Code Status above Command Line

Auto-fills the properties

Drop a Button under the Fader

- **Target is Console Button: button will be Select Last**
- **Resize rectangle to fit the text**

Drop a Rectangle to the left of the Select Last button

- **Target is Command**
- **Field 1 is Target Name, center justified, font size 10**
- **Type in the command box: Select_Last Sneak Enter**

Notice we can't type in the text to label it…it only shows the actual command. This also runs in the foreground. How do we run it in the background?

Drop a Label to the left of the Command button	
Change the text from Label to Sneak All	
Resize to fit all the text	
Change the Fill to red, and make the Label taller	
Target is Command	
Type in the Command Box: <U0> Clear Sneak 1 Enter	U0 is user 0 which runs in the background
Drop a Response IO Relay underneath to the left of the Command Label	
Drop a Foundation Motor to the left of the Relay	Auto-fills the properties
Resize to be the same size as the Relay	you will only see info on the motor if there is one connected
Drop a sACN input under Color Palette 9032	
Set the Target as Address 8/1	
With the sACN Input selected, create an Array from the Alignment tool	
Make sure Rectangle is selected	
Columns: 4 Spacing: 0 Rows: 1 Spacing: 0	
Select the Edit Mode Pointer in the tool bar	
Change the target to Address	
Change the start to 8/2 and renumber the 3 new Inputs	
Select all 4 sACN Inputs	
Ctrl+G to group	
Select the Freeform Polygon tool	
Draw a small spikey shape to the left of the sACN Inputs	
Double click to finish drawing	
Select the Image Library	Can also be imported from a USB
Select the Close Icon – ETC Cyan.png	
Drop the Close Icon under Color Palette 9032	
Resize the image to a small circle	
Select all the objects under the Color Palettes (from the Close Icon all the way to the Select Last button)	Remember the Color Palettes are grouped – don't include in the selection
Select the Alignment tool and Align Middle	
Distribute horizontally	

Oh, no! *With the sACN Inputs grouped, it doesn't distribute as expected.*

Manually rearrange to see all the objects. Make sure you keep the close icon where it is…we are going to add another object under Color Palette 9031!

Your Magic Sheet might look something like this:

Creating an Indicator Object using an Indicator Channel

Let's make a front light sub

In [Live], [1] [Thru] [10] [At] [Full] [Enter]	
[Select Last] [Record] [Sub] [101] [Label] Fronts [Enter]	
Without clearing your command line, [Time] [1] [Time] [Time] [1] [Enter]	Up Time 1, Down Time 1
{Hold} [Enter]	
[Clear] [Sneak] [Enter]	

Now let's add the sub to your Magic Sheet

Open the Magic Sheet Editor	
Select ⌈Group 102⌉	
Ctrl+C to copy, Ctrl+V to paste	
Change the target of the new object to Submaster with the ID 101	
Move to be directly below and aligned left with Color Palette 9031	

Test the Sub

Close the Editor, be in [Live]	
Press ⌈Fronts (S101)⌉	Notice how it takes 1 second to fade up and holds until you press it again

But we can't see the status on the Magic Sheet!

Only Color Palettes and channels change the state of the Magic Sheet object based on the console data.

Now let's make an indicator channel!

In Patch,	
Patch channel 9101 with no address as a MS Indicator	This is under show – already in the file. A copy of dimmer just renamed for clarity.
Label the channel Front Sub Indicator	
[Blind] [Sub] [101] [Enter]	
[9101] [At] [Full] [Enter]	Adds the channel at a level to the sub
Open the Magic Sheet Editor	
Select ⌈Fronts (S101)⌉	
Copy & paste a duplicate directly below the original	
On the original (top), change the Fill to No Fill (X)	
Select the copy (bottom)	
Change the Fill to a medium grey and Link to Channel Intensity	
Change the target to Channel 9101	
Remove all fields (set all to none)	
Select the Ordering tool and Send to Bottom	
Select the Sub and the Channel 9101 objects	
Align Top and Align Center	
Ctrl+G to group	
[Live]	
Hit ⌈Fronts (S101)⌉	Look at the indicator
Tap it again to turn it off	

A Review of Magic Sheet Settings

Gear Tab in Magic Sheet Editor

Select the Magic Sheets Settings tab	
Make sure interactive is checked (new to 3.0!)	Allows us to interact with the Magic Sheet rather than use it just as a visual aid

Knowing whether you are in Live or Blind is very important when doing fast programming or editing. Which is why there is such a big visual difference with background colors, title bars, command lines and even the word "Blind" in the top left corner of each monitor.

You can create the same sort of visual difference for your Magic Sheet.

Change the Live background to gradient	
Make the bottom color a gold	
Change the Blind background to gradient	
Make the bottom color a dark blue	
Close the editor	Look at the Magic Sheet in Live

Zoom to all…double tap with 2 fingers	
[301] [Full] [Full].	Cyc light
Now use the Magic Sheet, put Channel 301 in different colors.	
Watch the sACN inputs change for each address as the colors change	sACN objects are showing the Red, Green, Blue, and Amber values of channel 301)
⌈Select_Last Sneak Enter⌉	Only 301 sneaks; it is on the command line
⌈Sneak All⌉	Everything sneaks. It is not on the command line – remember user 0 in the command!

Your Magic Sheet might look something like this now:

A Review of Magic Sheet Displays & Navigation

Magic Sheet Settings

Click the gear on the bottom left hand corner near the tabs	to open the Magic Sheet Settings
Click Add View	to store a view while zoomed to all
Select [101]	
Zoom to selection	
Add another view	
Show how to navigate views…from the gear menu and a 3-finger swipe	
Save screenshot saves a screenshot to a USB	Great for printing off Magic Sheets for designers & gaffers
Select Magic Sheet Browser or tap once with 3 fingers	to open the MS Browser and select a different Magic Sheet
Lock	Locks the Magic Sheet so it cannot be zoomed or panned.
Limited Expanded Mode or press [Shift]&[Displays]	allows a Magic Sheet to be displayed at full screen

While in Limited Expand Mode, the monitor with the Magic Sheet will not display a tab number, command line, or status bar.

- Disable Keyboard Input - disables input from the face panel and an external keyboard. Fader and macro buttons will still work.
- Lock Fader Page - locks the fader page to the currently displayed one.

Magic Sheet List

Open a new tab, Magic Sheet List – Tab 14	

The Magic Sheet list displays a list of all created Magic Sheets, their labels, and how many views have been saved for each Magic Sheet.

It is easier to label & copy Magic Sheets here, especially while making many at once.

Popup Magic Sheet

This option allows you to view a favorite Magic Sheet in a small popup window over your current display.

A popup Magic Sheet can be assigned in Setup>User>Displays>Popup Magic Sheet. If a Magic Sheet has not been assigned in Setup, when you first click the popup Magic Sheet icon, you will be able to select a default Magic Sheet from the MS Browser strip

Setup > User > Displays > Popup Magic Sheet	

In Setup>User>Displays>Popup Nav Lock, you can enable or disable the zoom and scroll navigation for popup Magic Sheets.

The current popup is a bank of groups, so you always have groups at your fingertips.

Magic Sheet display Behavior

Magic Sheet tab behavior is set in the Magic Sheet Settings or Gear tab of the Magic Sheet Editor. There are three options:

- Normal Display – like all other Display Tabs (Silver tab label)
  - A locked tab will not be skipped.
  - A single click on the tab will draw focus to that tab.
- Channel Display - This mode uses the following rules:
  - When focus is drawn to the playback status display, a Magic Sheet channel display will be brought to the front.
  - Using **[Shift]&[Live]** cycles through the Magic Sheet channel displays as well as the Live channel displays.
  - Pressing **[Live]** or bringing a Live tab into focus will restore your last focused Magic Sheet channel display.
  - Magic Sheet channel displays in the locked frame will not be skipped when using the **[Tab]** key to cycle through tabs.
- Control – like other Control Tabs (Magenta tab label).
  - Locked tabs will be skipped.
  - A single click will not draw focus.
  - A double click will draw focus to the tab.

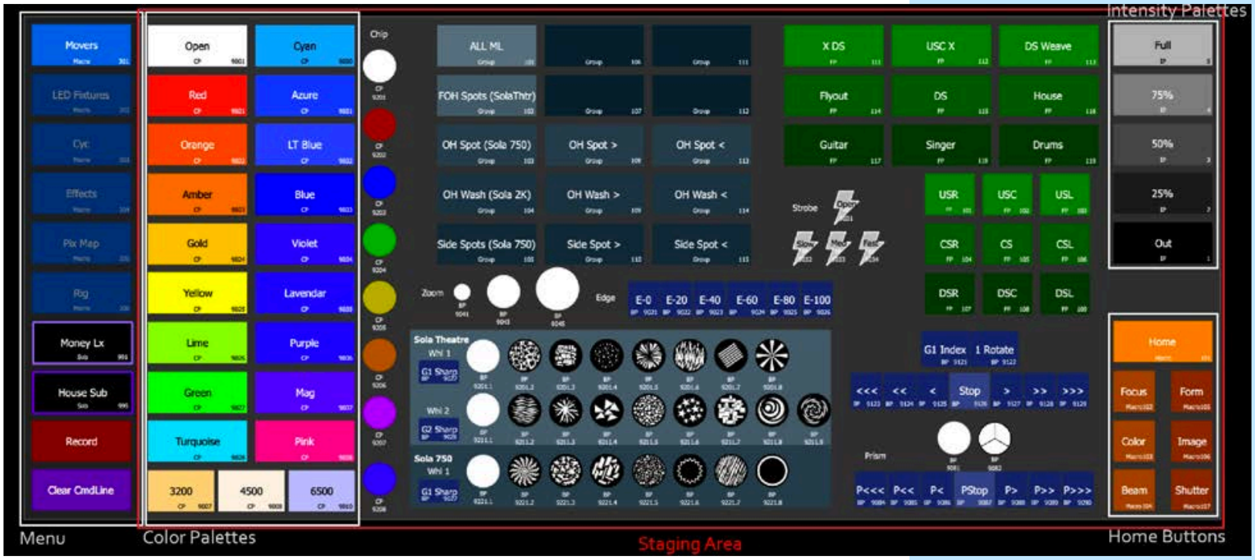
Tab Behavior

Look at Tab 1	Silver text – Normal or channel display
Next to Tab 2 on left hand monitor, open Tab 15, Submasters	Also, silver text – Normal or channel display
Next to Tab 15 on left hand monitor, add Tab 27, Color Picker	Magenta text – Control tab
Tab…Tab…Tab…	Follows tabs numerically
[Shift]&[Live]	Just navigates through Channel Displays
Open the Editor of Magic Sheet 102	
Click on the Gear Tab in the object library (Magic Sheet Settings)	
Notice behavior is Normal	follows normal display behavior
Change it to Control Display	
Close Editor	It's now purple
…hit the Tab name to bring focus to it	
Open the Editor	
Change it to a Channel Display	
Close the Editor	
[Shift]&[Live]	Now is selected as navigates through Channel Displays
Open the Editor and put it back to Normal	
Fire Snapshot 101	

Magic Sheet Show & Tell

Magic Sheet Examples

- A Staging area - what is copied to each Magic Sheet, so it looks basically the same.
- Look at Magic Sheet 301
  - Two gray boxes. Right hand large box is the "Staging area" – the background for all the content on each Magic Sheet
  - Left hand section is the Menu. It is on each of the Magic Sheets, laid out the same
    - o Macros to change to the other Magic Sheets (Tabs)
    - o Notice Movers (Macro 301) is brighter than the other macros to show which page we are on
    - o Two submasters (Money LX & House Sub)
      - ▪ Quick access to important looks you use a lot
      - ▪ Uses Indicator Channels
    - o A console button – Record
    - o A command box – Clear Command Line
    - o Targets on Magic Sheets with fixtures:
      - ▪ Rectangle Color Palettes on the left. They are linked to Target Color. They are the colors you would use a lot.
      - ▪ Intensity palettes on top right so you don't need the keyboard…with the palette's colors reflecting the intensity.
      - ▪ Bottom right are homing buttons – home all, then home individual categories



MOVERS - MAGIC SHEET 301

This is a Magic Sheet that a programmer would use to have easy access to targets pertaining to a system of lights. It doesn't show the fixtures, but it gives all the ML groups, CPs, BPs, FPs, IPs, and easy home buttons. You could choose to add the fixtures themselves if there is room. There often isn't.

- **Groups** – notice they are laid out visually to be easy for us to find what we need quickly. Gradients so that you can remember which color is with which fixture profile. Darkest boxes are blank groups for future recording. Brightest color is all the ML.
- **Color Palettes** – on the left. The squares are mixed palettes on all fixture Magic Sheets. The circles are color wheel palettes for the movers.
- **Focus Palettes** – laid out in a green gradient (green is the FP color in the console. Consistency!!) Underneath there is a small absolute, locked palette forming a positional starting point focus grid
- **Beam Palettes** – are shown visually when necessary, i.e. zoom size, strobe, gobos, prism. The rest are blue because BPs are blue in the console. The gobos are divided by fixture type (notice the background color matches the group color). The rotations lay out in speed order Left to Right

NOTE: *Consistency & Organization is important! It makes our lives easier as programmers, so you can easily find things. Future you and others stepping into your file will thank you later. Notice consistency of the color of the objects. By type palettes for all fixtures in the file are in the 9000s. Specific palettes & groups are in the 100s. This is the first Magic Sheet in the system. MS 301 & Macro 301.*

LED FIXTURES - MAGIC SHEET 302

This is very similar to the Movers Magic Sheet, but it is for the LED fixtures. Notice the CPs, IPs, and home macros are in the same place. We don't need Focus & Beam Palettes, so we have room to lay out the fixtures by system with Groups laid out to easily grab rows & columns & offsets

NOTE*:* *All the groups are in the 200s since this is the second Magic Sheet. MS 302 & macro 302*

CYC- MAGIC SHEET 303

This is very similar to the LED Fixtures Magic Sheet, but it is for the back wall. Notice the CPs, IPs, and home macros are in the same place. We don't need Focus & Beam Palettes, so we have room to lay out the cyc fixtures

- The cyc is multicell. We have placed each fixture's parent above its cells (you can autonumber by cell in Magic Sheets!) and provided groups for the parents, cells, and entire fixtures
  - Multicell grouping can get complicated, so laying out a lot of options while building the MS will save you a lot of time

  NOTE*:* *All the groups are in the 300s since this is the third Magic Sheet. MS 303 & macro 303*

- **Cyc Pixels**! The cyc pixels are in groups as well, in various offsets.
  - Click on the Pixel Wall popup. It looks like a popup, but it is just another copy of this Magic Sheet with the background color changed.
    - The Pixel Wall button is a macro to open that Magic Sheet. There is an expand icon so show that it is a "popup"
    - There is collapse button in the popup with a close icon that returns to the cyc MS
    - The pixels are laid out (using the array tool!) with groups to select rows, columns, and diagonals. Easy to make the groups with the pixels already laid out
  - Close the Pixel Wall popup back to the Cyc page

Effects – Magic Sheet 304

This is a target Magic Sheet with additional tools for the programmer.

- The Effects are laid out by parameter type and color coordinated (focus – green, beam – blue, intensity – orange: The colors of the palettes in direct selects.
- Either absolute Effects or relative Effects that can be applied to any fixtures in the associated palette or with the associated parameter.
- Color FX! There are a lot of them!
  - Relative color Effects are grey (the color of CPs in Direct Selects)
  - Two banks of Absolute color Effects that go from background to the Color Palette represented by the button. Notice the Effect numbers match the Color Palette number. Effects in the 9001s are background to open, CP 9001.
    - The first two rows are steppy absolute Effects. They have dwells of 0 and times of 0.5 The top row are Effects using a spread grouping. The bottom row are Effects using a grouping of 1. Notice how Effects in that row are all numbered ".1" Effects.
    - The second two rows are wave/fade absolute Effects. The have dwells of 0.5 and times of 0 The top row are Effects using a spread grouping. The bottom row are Effects using a grouping of 1.
    - "Special FX" in the top right are Effects with more than one parameter category, or Effects that are more specific

Open the Effects Magic Sheet (304)

Copy the Int fire Effect (962) and paste it outside the boundaries of the MS staging area.

How did we make this button?

- It is several layered objects grouped together, like the dummy channel as a sub indicator. Ungroup the button (Hint: Shift+Ctrl+G)
- The top object is the actual Intensity Palette. It has no fill so you can see through to the objects underneath
- The bottom object is a non-targeted rectangle of the same type, with the orange fill
- The "spikes" were made using the polygon tool and copying the shape several times in different colors, so they look like spikes of fire. They are grouped together and placed between the orange rectangle on the bottom and the Intensity Palette on top.

Close this Magic Sheet without deleting the copied objects!

Faders – Magic Sheet 304.1

On the FX Magic Sheets, look at the Control area in the bottom right

- This Magic Sheet has **7 virtual faders**. They are global FX faders filtered into parameter categories: focus, intensity, color, & beam. The faders are colored the same as the palette type's direct select color. There are also two unfiltered faders doing global size & rate
- **Home Faders** macro button (it homes the faders on this page) It is important if you are going to use motorized faders to have the ability to quickly home them, so you don't get stuck with an Effect doing something unexpectedly.
- **Stop FX** Buttons. They are color coded to the parameter type as well as a stop all button.
- **Faders Popup**. It opens MS 304.1, which looks like it pops up from that control page because of the way the popup button moves diagonally up left and becomes the collapse.
    - The background is the same color as the control box on the Effect page to emphasize the popup concept
    - There are large versions of the faders we already saw as well as a second bank of 10 empty faders. There are home fader buttons for each bank

Pixel Maps- Magic Sheet 305

This is a Magic Sheet pertaining to the pixels on the back wall

- It has the channels of the virtual media servers & layers in groups, laid out by the different maps. (notice they are in the 500s…this is the 5th Magic Sheet, MS 305 & Macro 305.)
- It has a grid of blank presets. When using pixel maps, presets (by layer or by the entire map) are a great way to insure you are storing the entire look of the map. Giving yourself this blank grid of presets gives you an easy place to store and recall the presets on the fly.

Rig- Magic Sheet 306

This is a visual systems Magic Sheet. It is one Magic Sheet that allows you to easily see what each system of multiple parameter fixtures are doing. The channels' fill property is linked to Target Intensity & Color.

- This is different than the Plot Magic Sheet. A Plot Magic Sheet would have the fixtures laid out on top of the plot or the scenery, with every fixture correctly spaced from each other regardless of type. However, it still shows spatial relationships between the fixtures in each individual system.
- This MS has CPs, IPs, and Home macros like the other MSs with fixtures on them.

Magic Sheet Navigation Tricks

Magic Sheet Navigation – In the Same Tab

We saw MS 301-307 were "tabbing Magic Sheets" – a series of Magic Sheets that look like they are tabs of the same sheet

Discuss Zoom to All function in making Magic Sheet Menus

- The background/framing of the Magic Sheet (the tabbing/Menu section and grey content "staging" area) are the same on all tabs so it automatically zooms to all and looks like it isn't moving

Open the Effects Magic Sheet (Macro 304)

Oh no! The view isn't right because we left an object outside of the "staging area" and they are included in zoom to all.

Open the Editor and delete the objects outside of the "staging area"

Close the Editor and lock the Magic Sheet

- Why does the example use "Macro" instead of "Magic Sheet" targets? Command Line interruption. Look at Macro List (Tab 18) to see that they are Background Macros.

Let's Create Magic Sheet 501

[Displays] {Magic Sheet} [501] [Enter]

Drop a rectangle and resize to create the "stage" — create empty space for content on each MS

Set a neutral color — This is your background

Drop a rectangle or a button and size/color how you like it — There will be 8 buttons, so size accordingly

Target – Macro 501

Fields Interior. Field 1 is Label, middle justified. — Make sure the text is easily readable

Field 2 is target name & field 3 is target ID. Put where you like them

We show the target small on each button so we can easily edit something if it is wrong without having to open the Magic Sheet to see the target.

Copy this button so you have 8 macro buttons (501-508) evenly spaced and aligned across the side of the staging space. Vertically or Horizontally. — You are making a menu/tabs.

Make the first menu button have an "indicator" state – brighter than the others – or have an outline. — Something to make it obvious we are on this page

{Magic Sheet} {Magic Sheet} to open the Magic Sheet List – Tab 14

Copy MS 501 to MS 502

Repeat until you have 8 Magic Sheets – 501 thru 508

Use Next and Last to label the Magic Sheets: 501: Home, 502: Moving Lights, 503: System, 504: Effects, 505: Target, 506: Programming, 507: Plot, 508: Control

Create Background Macro 501: Magic Sheet 501 ◆	Learn or open Macro editor (Macro Macro)
Copy to Macros 502 thru 508 and label them the same as Magic Sheets 501 thru 508	
Change the "indicator states" on each Magic Sheet so the correct macro is highlighted on each Magic Sheet.	
Test the navigation of your menu structure.	

Your Magic Sheet might look something like this:

Magic Sheet Navigation – In another tab/frame

This is when you use a Magic Sheet in one tab/frame to navigate displays & Magic Sheets in a different tab/frame.

Create a Tabs Only Magic Sheet

Open the Editor for Magic Sheet 501 – Home	
Select and copy the buttons Macros 501 – 508	
Create Magic Sheet 500 label Tabs	
Open the Editor	
Paste	
Change the target from Macro to Snapshot for all the buttons	
Autonumber from top to bottom until the buttons are numbered Snapshot 500.1 thru 500.8	

If you did vertical buttons:

Split the workspace (where MS was) into two columns and resize so the left frame is very narrow	Split side-by-side
Populate MS500 the narrow-left frame	
Populate MS501 in the large right frame	

If you did horizontal buttons:

Split the workspace (where MS was) into two rows and resize so the bottom frame is very narrow	Split top/bottom
Populate MS500 in the narrow frame	
Populate MS501 in the large frame	

Record Snapshot 500 label Tabbing	
Record Snapshot 500.1 – include only the tab with MS501 in it.	
Change tab with MS501 to Direct Selects - Effects	
Record Snapshot 500.4– include only the tab with DS in it.	
Using the tab MS, toggle between Snapshot 500.1 – Home & 500.4 - Effects	
Fire Snapshot 500	to restore to full screen

- Discuss how this could also be used as a menu to change one or more screens to different Magic Sheets.
- Could also use this to use direct selects instead of Magic Sheets for a target – like presets

MAGIC SHEET NAVIGATION – FULL SCREEN STARTUP

Make a Magic Sheet 599.1 Label Startup	
Drop the ETC Logo onto it	Under Images or Import from a USB
Copy that MS to 599.2 Label Press to Begin	
Make a "press to begin" button. Make sure it is within the bounds of the ETC logo object, and connects to Snapshot 500	No fields necessary

*(**Hint:** Use Text Object with a fill to change name without using Target Label).*

Reset all workspaces	
On the left touch screen, put MS 599.1.	
On the right touch screen, put MS 599.2	
Lock both.	

Idea was to then put them in LEM and record the Snapshot, but still doesn't work
In LEM, Currently only records one screen not both

Put both Magic Sheets in Limited Expand Mode	
Click the gear on the bottom left hand corner and record Snapshot 599	Only records 1 screen

Record Snapshot 599	
Make Background Macro 599 that calls Snapshot 599.	
In Setup > System > System Startup Macro, make it Macro 599.	

NOTE: *In order for a Snapshot to fire via startup macro on a multiconsole system, the Primary must be the first to come online. If the backup comes online first, it will not fire the macro*

Shut the console down	

Setting up a Home Tab

Turn Console On!

Snapshot 201 is fired from the startup macro, and we are in our startup Magic Sheets.

What is a Home Tab?

A Home tab/page is an informational page. It is the page you would default to with basic options or often needed things that would not require the user to go to more advanced Magic Sheet pages. This is a great page for operators. This page typically would not include the ability to edit data. Some examples are:

- Help popup if technical difficulties are occurring. They could have the installer's info or the programmer's info. ETC's info is a plus!
- Logos – show/project as well as installer
- Notes/instruction
- Fixture power on/off
- Start show macro
- Base look
- Cues (could also be a separate target page. May also use indicator channels to show when active.)
- Work light sub
- Departmental "popups": riggers, Stage Managers, electricians, or gaffers
- Status Indicators
- Small rig view if needed
- "Screen saver"

Home Tab – Info "Popup"

On MS501, import the Question Mark icon from the thumb drive, insert it into bottom right hand of the MS, target it to MS501.1, hide all fields.

Copy MS501 to MS501.1. Label "Help".

Edit MS501.1, remove all objects, including menu buttons

Change the fill of the "stage" to a dark cyan

Load the Close icon off the thumb drive and place it in the upper left-hand corner. Target it to MS501.

This will make it look like the help icon is popping up with the popup Magic Sheet

Load the ETC logo off the thumb drive, insert it in the center of the Magic Sheet. Make a text object underneath ETC contact info next to it.

Close and check your help "popup."

Home Tab – Individual Exercise

Add something to your home page.

You have the rest of the 30 minutes (should be about 15 minutes left).

You can add another popup for a specific department. You could add instructions for the day, start day cues or macros, work or house sub, etc.

It should be something each student can see themselves needing in their work environments.

We will review what each person has created with the rest of the class.

Setting up a System/Fixture Magic Sheet

System/Fixture Magic Sheets

System Magic Sheets are programming & informational Magic Sheets sorted by system or fixture type. They often contain all the information needed about the specific system...groups/channels for selection, associated targets and macros, and sometimes fixtures laid out linked to channel color & intensity. This allows you to stay on one page while working with a system, with everything at your fingertips. Some examples are:

- Tabs/pages are organized by system or fixture.
    - Moving Lights (all or by fixture)
    - LEDs (all or by fixture/system)
    - Cyc
    - Astera Tubes
- These pages often contain targets specifically for the system
    - Groups
    - Palettes & Presets
    - Macros
        - Selection
        - Query
        - Home
        - Record new targets
    - Effects

System Magic Sheet – Moving Lights

Open Magic Sheet 502 – Moving Lights	We are going to lay out all the movers per system.

FOH Fixtures

Drop a ML Spot. Link fill to target color & target intensity. Make Fields 2-4 Focus, Color, Beam	
Copy so there are 5 evenly spread fixtures in a row. From right to left, number 101 – 105	
Drop a group and resize to a rectangle. Renumber to Group 5 – FOH Movers. Put target name & target ID bottom interior justified. Make field 3 label and center justified. Place it to the left of the fixtures. Group this and all the fixtures together	

OH Spots

Copy one of the FOH Fixtures. Paste so there are 8 in 2 columns, 4 rows. Justify Fields 2-4 so they aren't overlapping.	
Renumber from right to left, then bottom to top, 111 - 118	
Copy the FOH group and change it to Group 108 – OH Spots. Center the group above the fixtures.	

OH Washes

Drop a ML Wash fixture and make it match all properties of the spots	Can also do Change Type
Lay out the same as the spots, with channels 121 - 128	
Make the group above the fixtures Group 109 – OH Wash	

Side Spots

Copy and Paste the OH Spots and their group

Rotate each column fixtures 90° so they are facing in towards each other (><) and justify the fields 2-4 so they are not overlapping

Change the group to 105 – Side spots.

Color Palettes

Group each system of lights with their group and arrange them so you have room for some palettes.

Drop a Color Palette, appropriately size, and target 9021.

Put target name & target ID bottom interior justified.

Make field 3 label and center justified. Link Fill to Target color.

Starting with CP 9022, make an array that is 2x9 (either horizontally or vertically depending on the layout)

There should be CPs 9021 – 9038

Close the Editor.

What's missing? Open! It is important to always have an open palette.

Copy a palette and paste it in line with the palette array. Resize so it takes up both rows. Make it CP 9001 – Open.

Focus Palettes

Create a 3x3 Focus Palette array with Focus Palettes 101 thru 109.

They should all have the label center and the target name & number in the interior of the button.

Change the fill color to green to match the color Focus Palettes appear in the console.

Intensity Palettes

Drop an Intensity Palette button. You can make it smaller – even half the width of the Color Palette. It should have the label center and the target name & number in the interior.

Copy & paste so there are 5 in a row numbered 1-5.

IP1 is Out. IP5 is Full. Change the fill color for each button to reflect the intensity of the palette

Additional buttons

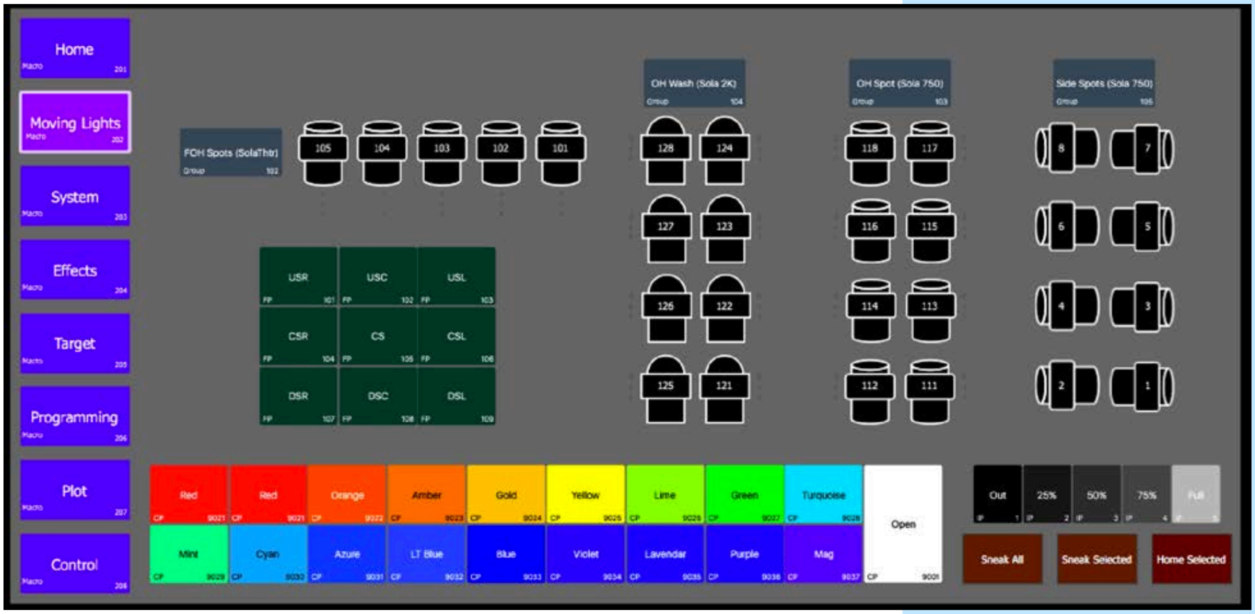
Create 3 buttons: one that homes selected, one that sneaks selected, and one that sneaks all.

Your choice as to how to do this.

One solution: *3 command buttons made with labels: Home Enter, Sneak Enter, Clear_CmdLine Sneak Enter*

Arrange all the new palettes and buttons so they lay out nicely with the fixtures

Your Magic Sheet might look something like this:



SYSTEM MAGIC SHEET – INDIVIDUAL EXERCISE

Students can choose to continue working on this Magic Sheet (maybe layout Beam Palettes or more macros) or use the blank system Magic Sheet 203 to work on a different system of fixtures. They have 15 minutes to play.

Setting up a “Target” Magic Sheet

Target Magic Sheets

Target Magic Sheets are programming Magic Sheets sorted by target type. They often contain targets that can be used for all fixtures in the rig. This Magic Sheet is like using direct selects, but you can organize in a way that makes sense to you and the show. Programmers will often divide the page by fixture type. Another common trick is to make the most used targets more prominent than all the others. Helpful trick. Divide a screen with one frame group direct selects, other frame tabbing target MS system. Some examples are:

- Tabs/pages are organized by target type.
  - Effects (this is one of the most common pages – MS 304 in this file)
  - Cues
  - Submasters
  - Groups
  - Color Palettes
  - Beam Palettes
  - Focus Palettes
  - Presets
  - Shutters

Target Magic Sheet – Effects

Open Magic Sheet 504 – Effects.

Let’s lay out effects by parameter type, and color code them according to their direct select color to keep it consistent and easy.

Intensity Effects

Drop an Intensity Palette. Size it appropriately. Change the target to Effect. Change field 3 to the label in the center and justify target name and ID on the bottom.

Create an array of 3 columns, 5 rows. Renumber the columns as:

Column 1: FX 931 – 935

Column 2: FX 941 – 945

Column 3: FX 951 – 955

Drop a label with the text Linear under the first column

Drop a label with the text Absolute centered under the remaining two columns

Drop a larger label with the text Intensity centered above the array

Focus Effects

Drop a Focus Palette. Change it to Effect 900 and make it the same size and field properties as the Intensity effect.

Using the palette object assures the right color.

Change the Focus Palette start to 901. Create an array of 2 columns, 5 rows.

Copy the Intensity label and change it to Focus. Center above the array.

Beam Effects

Drop a Beam Palette. Change it to Effect and make it the same size and field properties as the intensity effect.

Using the palette object assures the right color.

Create an array of 2 columns, 2 rows. Renumber the rows as:

Row 1: FX 922.1 + 922.2

Row 2: FX 921.1 + 921.2

Copy the Intensity label and change it to Beam. Center above the array.

Color Effects

This will take the longest. If short on time this can be shortened or left to them to choose to do in the individual exercise. Or the fill color can be left grey and field 3 can be a centered label – which will say the name of the color

Drop in a button and resize to be the same width as the other effects, but half the height.

There will be 13 in column. Make sure it is still easy to pick with your finger.

Make it Effect 9001, and have fields 1 & 2 be target name & ID. Leave them centered

Create an array of 1 column, 13 Rows. Autonumber as:

§ **9001: (Open)**
§ **9021: (Red)**
§ **9022: (Orange)**
§ **9023: (Amber)**
9025: (Yellow)
§ **9027: (Green)**
§ **9030: (Cyan)**
§ **9032: (Lt Blue)**
§ **9033: (Blue)**
§ **9035: (Lavender)**
§ **9036: (Purple)**
§ **9037: (Magenta)**
§ **9038: (Pink)**

Remember: to skip numbers, keep clicking the same button until it is numbered correctly

Change the fill colors to match the Color Palette. Group them all and copy them to the right

Ungroup and renumber them as the 0.2 version of the effect (i.e. 9001 becomes 9001.2)

Drop a label with the text Step under the first column

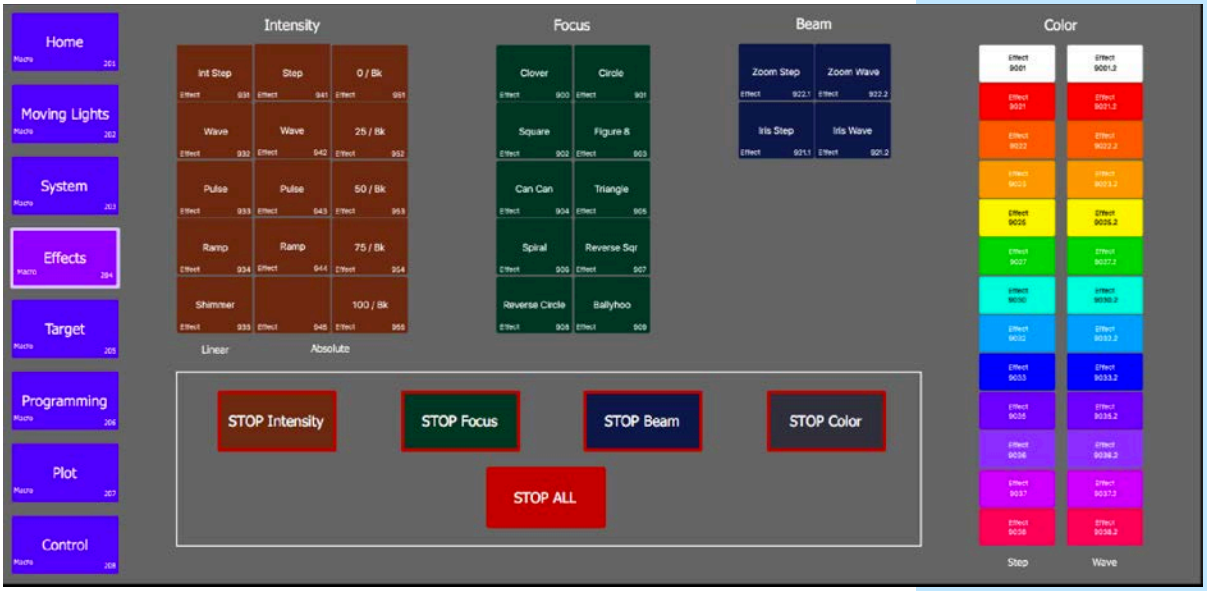
Drop a label with the text Wave under the second

Copy the Intensity label and change it to Color. Center above the array.

Arrange all the effects to lay out nicely

The stop effects were not made in the class exercises but are included in the file as an example, and strongly recommended for the students to make as part of their individual exercise. If time, they could be made as a class before moving on to the exercise.

Your Magic Sheet might look something like this:



TARGET MAGIC SHEET – INDIVIDUAL EXERCISE

Students can choose to keep working on this effects page… maybe add in the stop effects buttons. Or they can work on the blank "Target" Magic Sheet 205 to lay out a different target.

Setting up a Plot Magic Sheet

What is a Plot Magic Sheet?

Plot Magic Sheets give a visual aid of the fixtures in the show. They include a layout of the fixtures and show what they are doing by linking the fixtures to channel color & intensity. Often, a programmer will import the plot or scenery as the background and layout the fixtures on top. Some people then remove the image as the background once the fixtures are laid out. There are many ways to organize plot Magic Sheets.

Some examples are:

- Overall Plot:
- By Scenery
- By Studio/Set (TV/Film)
- By Rig…lays out fixtures visually by system rather than location (MS 306.)

Plot Magic Sheets often function as programming Magic Sheets as well. These include tools for programming the fixtures as well as the plot layout. This is quite common in television studios that function out of Magic Sheets

- Groups
- Palettes
- Macros, such as:
  - Sneak
  - + & - %
  - Record/Update to specific targets

Plot Magic Sheet – Create a Plot

Open Magic Sheet 507 – Plot	
Select the background/staging box and change it to a grey outline with no fill	
Group all the content	
Change the Live Background to Image	
Import the Light Plot from the USB drive.	
Resize so it fits inside the staging area	
Move the Magic Sheet content until it lays on top of the plot nicely.	Okay to move the Magic Sheet if the objects stays in the same size relative to each other, because of the zoom to all function.
change the Blind Background to the image as well and make sure it is the same size, so it looks the same in blind	Even though we can't see the blind background while editing
Drop truss onto the Magic Sheet. Resize so it is the size of the FOH truss. Copy it onto the 1st through 4th electrics and the cyc top/bottom.	

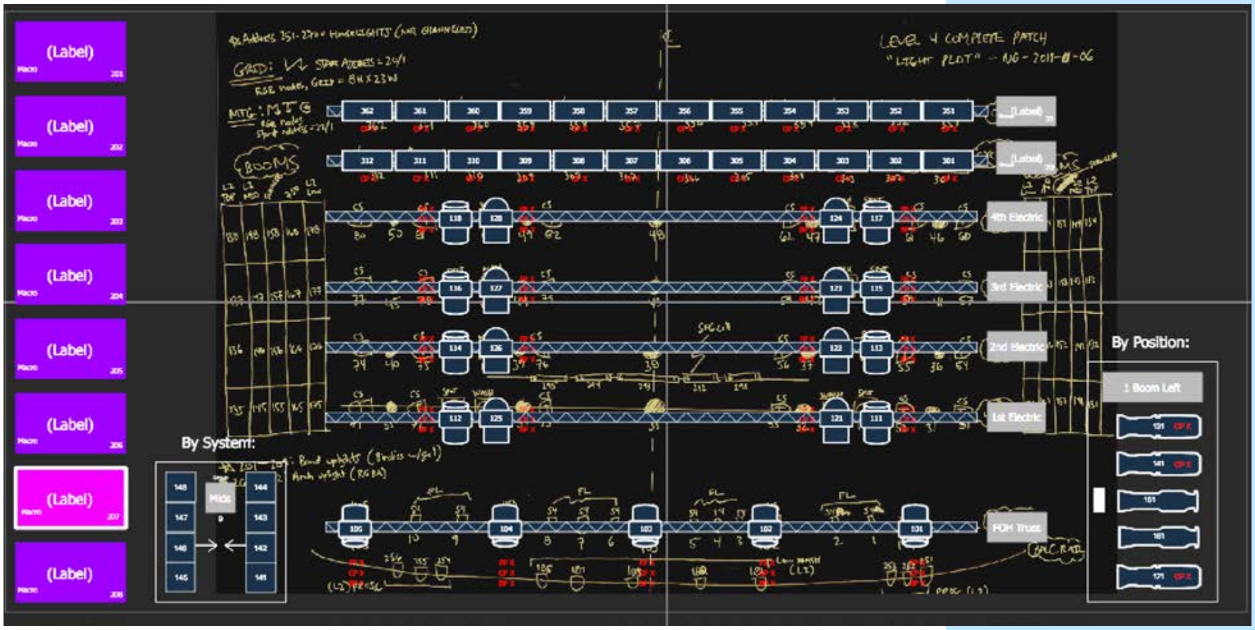
Drop a Moving Spot onto the Magic Sheet.	
Link Fill to Target Color & Intensity. Fields should be exterior bound. **Field 1 center justified is Target ID. Field 2 is Focus. Field 3 is Color. Field 4 is Beam. Fields 2 – 4 should be bottom center.**	Laying out FCB as it appears in live summary.
Rotate the fixture 180° so it is pointing up.	
Move it on top of one of the drawn FOH spot fixtures. Make sure the size fits the truss	
Copy and paste and until all spots are placed. (5 FOH, 8 Overstage)	
Autonumber to correctly channel the fixtures per the plot	
Rotate the Overstage fixtures 180° so are pointing down.	Notice the Overstage fixtures don't lay out nicely – the fields overlap each other
Change justifications for fields 2-4 for the Overstage fixtures so they are on the off-stage sides	

Drop a rectangle onto the Magic Sheet	
Resize so it is roughly the size of one of the drawn cyc top fixtures.	Not using a stock fixture because they can't be non-proportionally resized.
Link fill to channel color & intensity. Make the line white	
Copy the cyc fixture until there are 12 (you can use an array) and line them up with the drawn fixtures	
Autonumber to match the cyc top	
Copy all twelve and paste to create the bottom cyc	Hint – if you make the start channel in edit mode 351, it will number correctly

Drop a Label and make the Fill gray. Resize so you see a box around the label. Label it FOH truss and put it next to the FOH truss	
Label all Electrics	
Cyc Top & Bottom Labels should turn into groups. Cyc top is group 20 and cyc bottom is group 21. Make sure we have the target name & ID somewhere. You can also remove the label text and make field 3 label.	
Close the Editor.	This looks messy with plot image in the background.
Open the Editor and change live background back to solid	
Check 'Use Blind Background while editing'	so can still lay out on top of the plot image

The By Position and By System fixtures such as the Overhead Wash fixtures were not made in the class exercises but included in the file as an example of each

Your Magic Sheet might look something like this:



Plot Magic Sheet – Individual Exercise

Continue to lay out the plot as you would like.

You may want to lay out the spot fixtures.

You may want to lay out on the side the booms either by position or by system (Is there room?)

You can also put down targets you might want to use with individual fixtures such as palettes and presets.

You can create a bank of groups to put on the side of difficult to lay out systems you may not want individual fixtures for (Side systems, Front light, Back wall pixels, High Sides, etc.).

You may also want some macros or commands that home or sneak if you are using palettes.

Programming Surface Magic Sheet

What is a Programming Surface Magic Sheet?

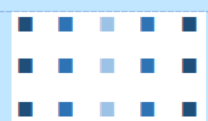
We have been talking about Magic Sheets used for programming. Any Magic Sheet that includes the means for changing what a fixture is doing could be considered programming Magic Sheets. However, Programming Surface Magic Sheets are more specifically for programmers. They are organized to streamline a programmer's workflow. It is a page that puts the programmer's commonly used tools directly at their fingertips. Some examples are:

Fire Macro 1	Should stop all effects
Fire Snapshot 102	Check out the upper right-hand screen

What is this page? This is Magic Sheet that is just an array of macros (8001-8075…great for merging.), colored in a gradient as a visual aid. This contains some commonly used macros programming. Blank spaces are empty macros so you can create a new macro as you program and have it already at your fingertips.

- Command Line – See what the LD is doing!
- Timecode clock
- Navigation – Pulling up other tabs such as flex Live/blind channels, PSDs, Magic Sheets, or direct selects in a predetermined tab.
- Faders (MS 304.1) – great for busking
- "Widgets" – tools for easily manipulating a complicated parameter

Create a Programmer Magic Sheet

Open Magic Sheet 506 – Programming	
Drop in a Rounded Rectangle. Color it something dark. Resize so it is can easily be pressed by a finger. Make it Macro 2001. Label in the Middle. Target Name & target number should both be interior fields – arrange as wanted.	
Make it so the next created macro button will be 2002 (edit mode)	
Create an array with macro 2001 that is 5x10, no spacing – whichever orientation is fine.	
In the longer direction, change the color of each row or column so that it is lighter than the previous, creating a mirrored in gradient for ease of finding macros	
In the Macro editor, create macros 2001 through 2050.	

There does not need to be any content. Ahead of programming a gig it makes sense to create macros that lay out nicely in this Magic Sheet. You can also leave some blank so if you find you need something while programming, there is already a place for it.

Programmer Magic Sheet - Individual Exercise

You have 20 minutes (or however long is left in the 30-minute unit) to create a shutter "widget."

It should have the ability to easily select each frame thrust, frame angle, and shutter assembly as well as home all thrust and all angle.

It could also have buttons that does the pairs of thrusts (A&C, B&D) as well as Beam Palettes for boxes (the show file has Beam Palettes the students can use direct selects to look at).

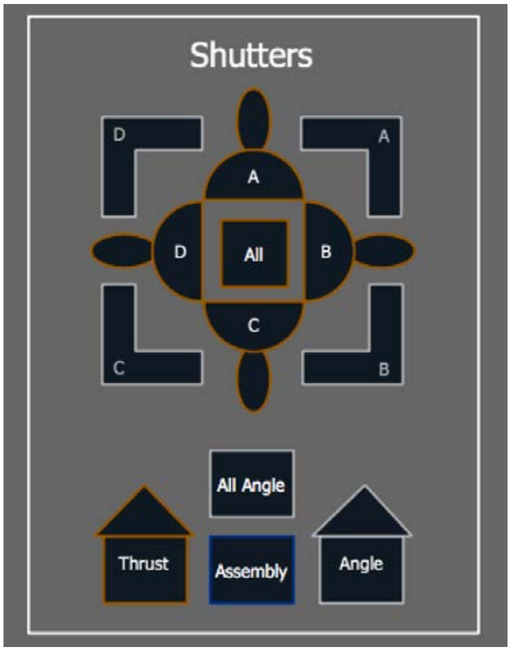
Hint! Use commands! The commands for selecting shutter parameters:

Frame_Thrust_A

Frame_Angle_A

Frame_Assembly

An Example of a Shutter Tool from completed class file:



If this tool is not applicable to a student, they may choose to make something else that they can justify being on a programmer Magic Sheet.

Control Tab Magic Sheet

What is a Control Tab Magic Sheet?

Control Tabs are for functions that are more maintenance rather than programming. They often affect the overall system. It is a safe place to put potentially disruptive actions that you wouldn't want to accidentally do during normal programming. Some examples are:

- Fixture power on/off
- Channel Check
- System Setting Macros
  - Enable/Disable
  - Sneak Time
  - Manual Time
- Status Indicators
- Lamp Functions
- sACN Output
- Maintenance

Control Tab:

Using Color Palettes and Copy To for Magic Sheet Object Status

Use Remote On / Remote Off examples with point targets and indicator channel. Macros will be used to set the color of channel 9290 in CP 290, which will be on the Magic Sheet to show the status of RFR Remotes. We use Color Palettes, so we don't have indicator channels with values contributing to cues and subs.

Patch Channel 9290 as a Generic RGB LED

In Blind, make Color Palettes CP 290, CP290.1 and CP290.2

CP 290 – Label RFR Status

CP 290.1 – Label On

Make Channel 9290 Green

CP 290.2 – Label Off

Make Channel 9290 Red

Make Background Macros 291 and 292

Macro 291 – RFR Enable. Macro should read:

RFR enable ◆ Blind Color_Palette 290 ◆ 9290 Recall_From Color_Palette 290.1 ◆

Macro 292 – RFR Disable. Macro should read:

RFR disable ◆ Blind Color_Palette 290 ◆ 9290 Recall_From Color_Palette 290.2 ◆

Put CP 290 linked to channel color and both macros on Magic Sheet 508

Control Tab:

Using "Cascading" macros to make status and task be toggle-able

Use Remote On / Remote Off examples and show a cascading (or auto-moving) macro system. This system creates a toggle by moving the macros so the fired one is opposite each time.

Macro 290 – RFR Toggle. Macro Reads:

Macro_Button 291◆ Macro 291 Copy_To Copy_to 293 ◆ Macro 292 Thru 293 Copy_To Copy_to 291 ◆◆

Delete the macro buttons on the control tab

Copy the Color Palette and change the copy to Macro 290 with a clear fill. Align Center & Middle & Group

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